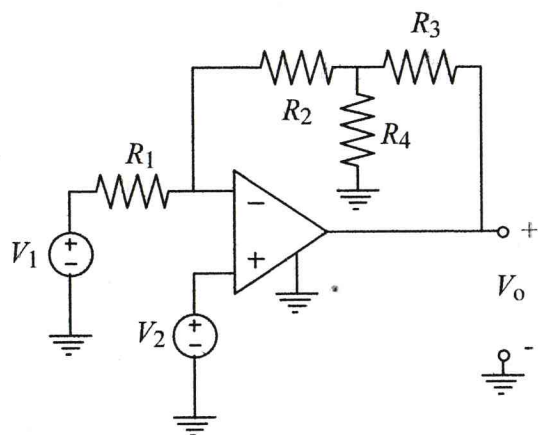


Esercizio 1.1)

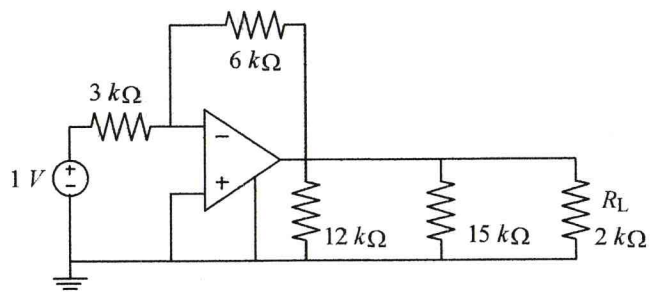
Determinare V_o .



$$[V_o = \frac{(R_1 + R_2)(R_3 + R_4) + R_3 R_4}{R_1 R_4} V_2 - \frac{R_2(R_3 + R_4) + R_3 R_4}{R_1 R_4} V_1]$$

Esercizio 1.2)

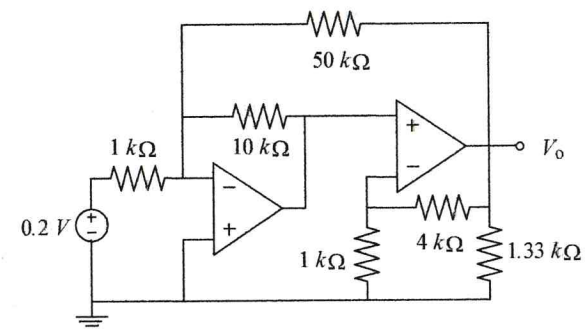
Determinare la potenza assorbita dal resistore R_L .



[2 mW]

Esercizio 1.3)

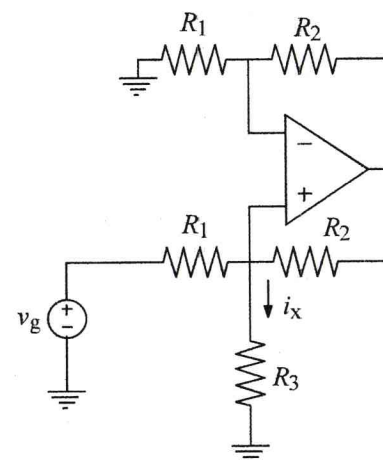
Determinare il valore di V_o .



[- 5 V]

Esercizio 1.4)

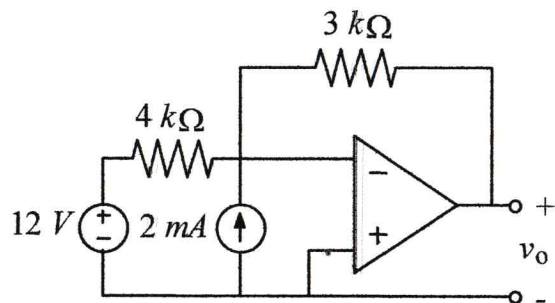
Determinare l'espressione della corrente i_x .



$$[i_x = \frac{v_g}{R_1}]$$

Esercizio 1.5)

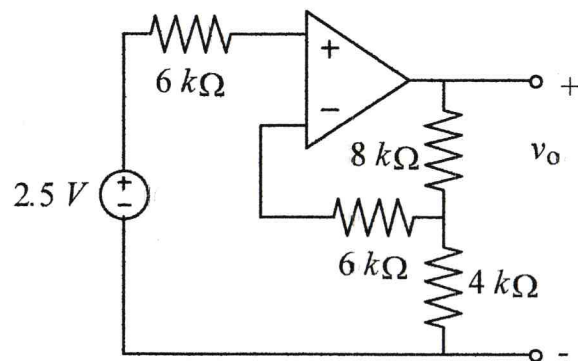
Determinare il valore di v_o .



[- 15 V]

Esercizio 1.6)

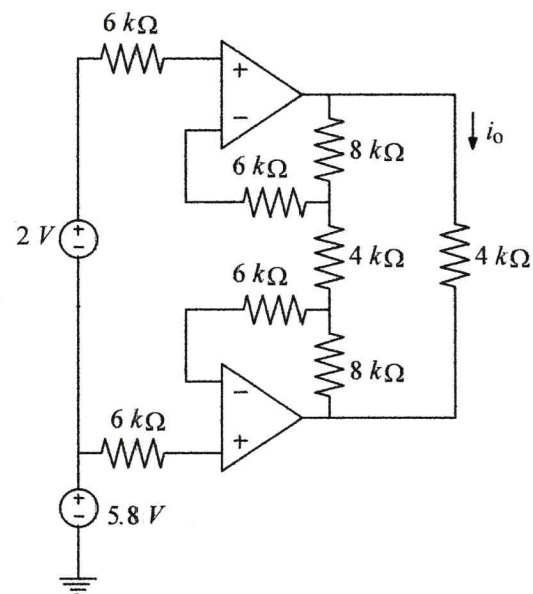
Determinare il valore di v_o .



[7.5 V]

Esercizio 1.7)

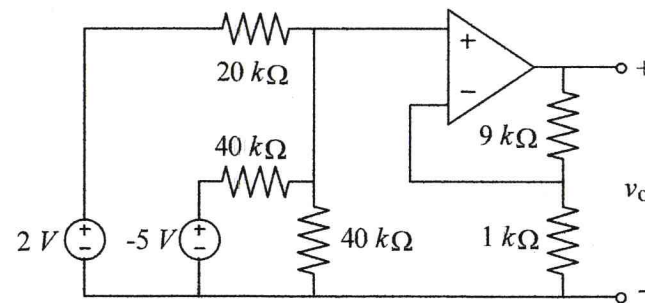
Determinare il valore di i_o .



[2.5 mA]

Esercizio 1.8)

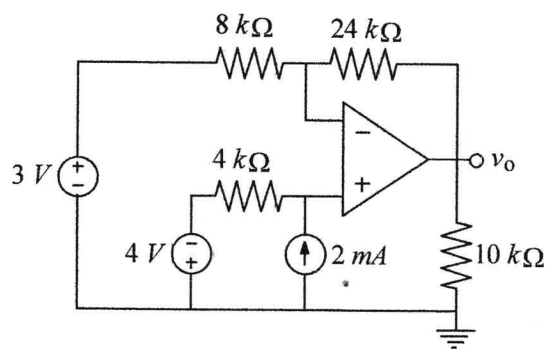
Determinare il valore di v_o .



[-2.5 V]

Esercizio X.9)

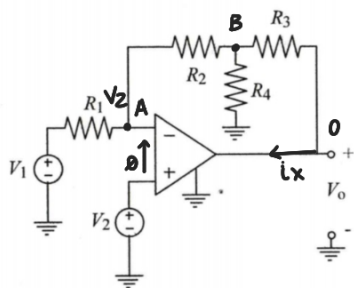
Determinare v_o .



[7V]

Esercizio 1.1

Determinare V_o .



$$V_o = \frac{(R_1 + R_2)(R_3 + R_4) + R_3 R_4}{R_1 R_4} V_2 - \frac{R_2(R_3 + R_4) + R_3 R_4}{R_1 R_4} V_1$$

$$V_A = V_2$$

$$A) \frac{V_1}{R_1} + \frac{V_B - V_A}{R_2} = 0 \rightarrow V_1 R_2 + V_B R_1 - V_2 R_1 = 0$$

$$B) \frac{V_A - V_B}{R_2} - \frac{V_B}{R_4} + \frac{V_o - V_B}{R_3} = 0$$

$$C) -i_x + \frac{V_B - V_o}{R_3} = 0 \rightarrow i_x = \frac{V_B - V_o}{R_3}$$

$$\rightarrow V_B = \frac{V_2 R_1 - V_1 R_2}{R_1}$$

$$\rightarrow \frac{V_o - V_B}{R_3} = -\frac{V_A - V_B}{R_2} + \frac{V_B}{R_4} \rightarrow V_o - V_B = -\frac{R_3}{R_2} V_A + \frac{R_3}{R_2} V_B + \frac{R_3}{R_4} V_B$$

$$V_o = -\frac{R_3}{R_2} V_A + \frac{R_3}{R_2} V_B + V_B + \frac{R_3}{R_4} V_B$$

$$\rightarrow V_o = -\frac{R_3}{R_2} V_2 + \frac{R_3}{R_2} \left(\frac{V_2 R_1 - V_1 R_2}{R_1} \right) + \frac{V_2 R_1 - V_1 R_2}{R_1} + \frac{(R_3) V_2 R_1 - V_1 R_2}{R_1 R_4}$$

$$V_o = -\frac{R_3}{R_2} V_2 + \frac{V_2 R_1 R_3}{R_2 R_1} - \frac{V_1 R_2 R_3}{R_2 R_1} + V_2 - \frac{V_1 R_2}{R_1} + \frac{V_2 R_1 R_3}{R_1 R_4} - \frac{V_1 R_2 R_3}{R_1 R_4}$$

$$V_o = -\frac{R_3}{R_2} V_2 + \frac{R_3}{R_2} V_2 - \frac{R_3}{R_1} V_1 + V_2 - \frac{R_2}{R_1} V_1 + \frac{V_2 R_3}{R_4} - \frac{R_2 R_3}{R_1 R_4} V_1$$

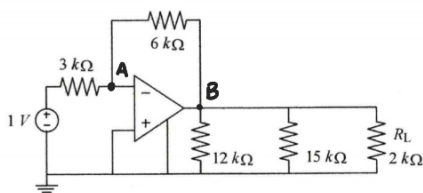
$$V_o = \frac{-R_3 R_4 + R_3 R_4 + R_2 R_4 + R_2 R_3}{R_2 R_4} V_2 - \frac{R_3 R_4 + R_2 R_4 + R_2 R_3}{R_1 R_4} V_1 =$$

$$= \frac{R_2 (R_4 + R_3)}{R_2 R_4} V_2 - \frac{R_2 (R_4 + R_3) + R_3 R_4}{R_1 R_4} V_1 =$$

$$= \frac{R_4 + R_3}{R_4} V_2 - \frac{R_2 (R_4 + R_3) + R_3 R_4}{R_1 R_4} V_1$$

Esercizio 1.2

Determinare la potenza assorbita dal resistore R_L .



[2 mW]

$$[V][G] = [I] \rightarrow [V] = [G]^{-1}[I]$$

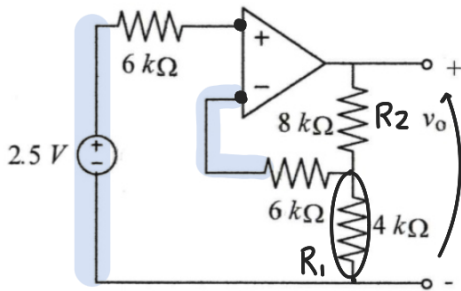
$$\begin{bmatrix} \frac{1}{2k} & -\frac{1}{6k} \\ -\frac{1}{6k} & \frac{49}{60k} \end{bmatrix} \begin{bmatrix} 0 \\ V_B \end{bmatrix} = \begin{bmatrix} \frac{1}{3k} \\ 0 \end{bmatrix}$$

$$\begin{cases} -\frac{1}{6k} V_B = \frac{1}{3k} \\ \frac{49}{60k} V_B = 0 \end{cases} \rightarrow V_B = -2 \text{ V}$$

$$P = R i^2 = \frac{V^2}{R} = \frac{4}{2} = 2 \text{ mW}$$

Esercizio 1.6

Determinare il valore di v_o .



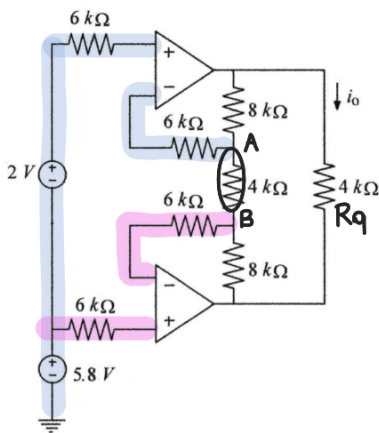
per partitore di tensione :

$$V_x = \frac{R_1}{R_1 + R_2} V_o \rightarrow V_o = \frac{2,5 \cdot (8+4)}{4} = 7,5 \text{ V}$$

[7.5 V]

Esercizio 1.7

Determinare il valore di i_o .



$$V_1 = 2 + 5,8 = 7,8 \text{ V}$$

$$V_2 = 5,8 \text{ V}$$

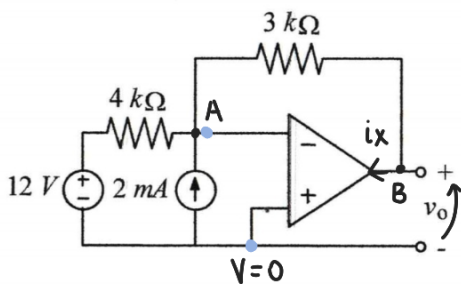
$$V_A - V_B = 2 \text{ V}$$

$$\text{per partitore : } V_{AB} = \frac{4}{4+8+8} \cdot V_{TOT} \rightarrow V_{TOT} = 10 \text{ V}$$

$$V_{TOT} = R_q \cdot i_o \rightarrow i_o = \frac{V_{TOT}}{R_q} = \frac{10 \text{ V}}{4 \text{ k}\Omega} = 2,5 \text{ mA}$$

Esercizio 1.5

Determinare il valore di v_o .



[-15 V]

$$[V][G] = [I]$$

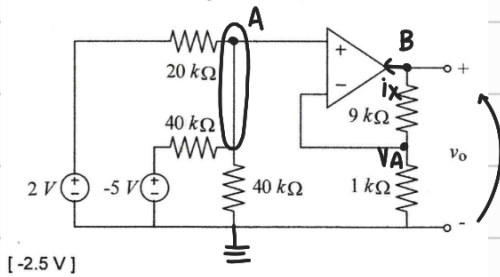
$$\begin{bmatrix} \frac{7}{12} \text{ m} & -\frac{1}{3} \text{ m} \\ -\frac{1}{3} \text{ m} & \frac{1}{3} \text{ m} \end{bmatrix} \begin{bmatrix} V_A \\ V_B \end{bmatrix} = \begin{bmatrix} 3 \text{ mA} + 2 \text{ mA} \\ -i_x \end{bmatrix}$$

$$V_A = 0 \quad V_B - V = V_B = V_o$$

$$-\frac{1}{3} \text{ m} V_B = 3 \text{ m} + 2 \text{ m} \rightarrow -\frac{1}{3} V_o = 5 \rightarrow V_o = -15 \text{ V}$$

Esercizio 1.8

Determinare il valore di v_o .



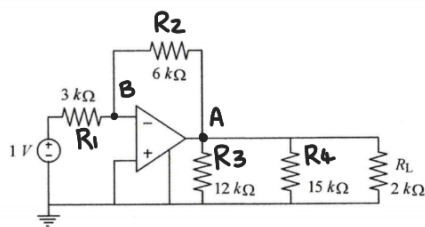
- possiamo applicare Millman sul nodo A

$$\text{Applichiamo Millman: } V_A = \frac{\frac{2}{20} \text{ m} - \frac{5}{40} \text{ m}}{\frac{1}{20} + \frac{1}{40} + \frac{1}{40}} = \frac{-\frac{1}{40}}{\frac{4}{40}} = -\frac{1}{4} \text{ V}$$

$$\text{per il partitore: } V_A = \frac{1 \text{ k}}{9 \text{ k} + 1 \text{ k}} V_o \rightarrow V_o = 10 V_A = -2,5 \text{ V}$$

Esercizio 1.2

Determinare la potenza assorbita dal resistore R_L .



[2 mW]

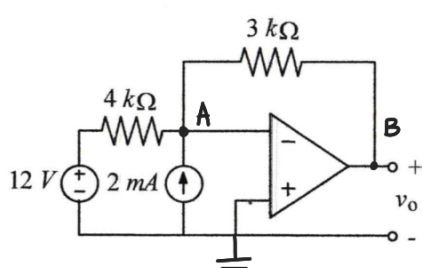
$$\begin{cases} -i_x - \frac{V_A}{R_3} - \frac{V_A}{R_4} - \frac{V_A}{R_L} + \frac{V_B - V_A}{R_2} = 0 \\ \frac{V_A - V_B}{R_2} + \frac{1}{R_1} = 0 \end{cases}$$

$$\rightarrow \frac{V_A}{R_2} - \frac{V_B}{R_2} + \frac{1}{R_1} = 0 \rightarrow \frac{V_A}{6} = -\frac{1}{3} \rightarrow V_A = -2 \text{ mV}$$

$$P = V i = \frac{V^2}{R} = 2 \text{ mW}$$

Esercizio 1.5

Determinare il valore di v_o .



[-15 V]

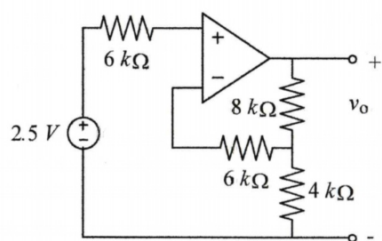
$$\begin{cases} I + \frac{V}{R_1} + \frac{V_B - V_A}{R_2} = 0 \\ \frac{V_A - V_B}{R_2} - i_x = 0 \end{cases}$$

$$\rightarrow 2 + \frac{12}{4} + \frac{V_B}{3} = 0 \rightarrow 2 + 3 + \frac{V_B}{3} = 0$$

$$\rightarrow V_B = -15 \text{ V}$$

Esercizio 1.6 :

Determinare il valore di v_o .



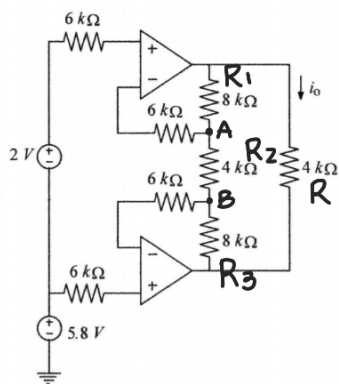
[7.5 V]

per partitore di tensione :

$$V_x = \frac{R_1}{R_1 + R_2} V_o \rightarrow V_o = \frac{2.5 \cdot (8+4)}{4} = 7.5 \text{ V}$$

Esercizio 1.7

Determinare il valore di i_o .



[2.5 mA]

$$V_A = 5.8 \text{ V} + 2 \text{ V} = 7.8 \text{ V}$$

$$V_B = 5.8 \text{ V}$$

$$V_A - V_B = 7.8 \text{ V} - 5.8 \text{ V} = 2 \text{ V}$$

per il partitore di tensione:

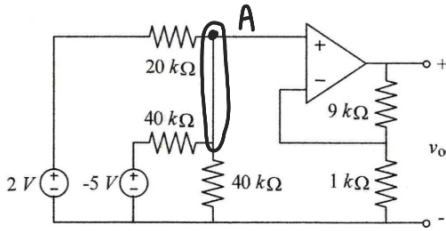
$$V_{AB} = \frac{R_2}{R_2 + R_1 + R_3} V_{TOT}$$

$$\rightarrow V_{TOT} = V_{AB} \cdot \frac{20V}{4V} = 2 \cdot \frac{20V}{4V} = 10V$$

$$I_0 = \frac{V_{TOT}}{R} = \frac{10V}{4k\Omega} = 2.5mA$$

Esercizio 1.8

Determinare il valore di v_o .



[-2.5V]

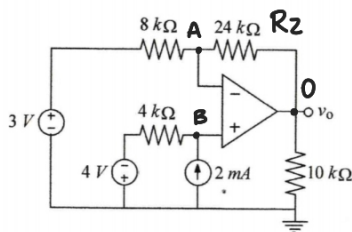
- possiamo applicare Millman sul nodo A

Applichiamo Millman:
$$V_A = \frac{\frac{2}{20}m - \frac{5}{40}m}{\frac{1}{20} + \frac{1}{40} + \frac{1}{40}} = \frac{-\frac{1}{40}}{\frac{1}{20}} = -\frac{1}{4}V$$

per il partitore:
$$V_A = \frac{1k}{9k+1k} V_o \rightarrow V_o = 10 V_A = -2,5V$$

Esercizio 1.9

Determinare v_o .



[7V]

Per Millman:

$$V_B = \frac{-\frac{4}{4} + 2}{\frac{1}{4}} = 4V \quad V_B = V_A$$

Applico metodo nodi sul nodo A:

$$\frac{V - V_A}{R_1} + \frac{V_o - V_A}{R_2} = 0 \rightarrow \frac{3-4}{8} + \frac{V_o-4}{24} = 0$$

$$\rightarrow -3 + V_o - 4 = 0 \rightarrow V_o = 7V$$